

Milestone 2: Dataset and Problem Approval Memo

Team ML Musketeers (Group 21) - CSCI 3052U, Fall 2026 - companion files: *notebooks/01_m2_edu.ipynb*, *src/data.py*, *data/split_indices.json*

1. Problem and task

Task: 3-class supervised classification. **Input:** a student's enrollment record plus academic results for semester 1 (early set, 30 features) or semesters 1 and 2 (full set, 36 features). **Output:** final status - Dropout, Enrolled or Graduate. **Users:** academic advisors and retention staff who want to prioritize outreach. **Research questions and success criterion** are unchanged from the Milestone 1 proposal (see Section 8 for the one clarification).

2. Data card

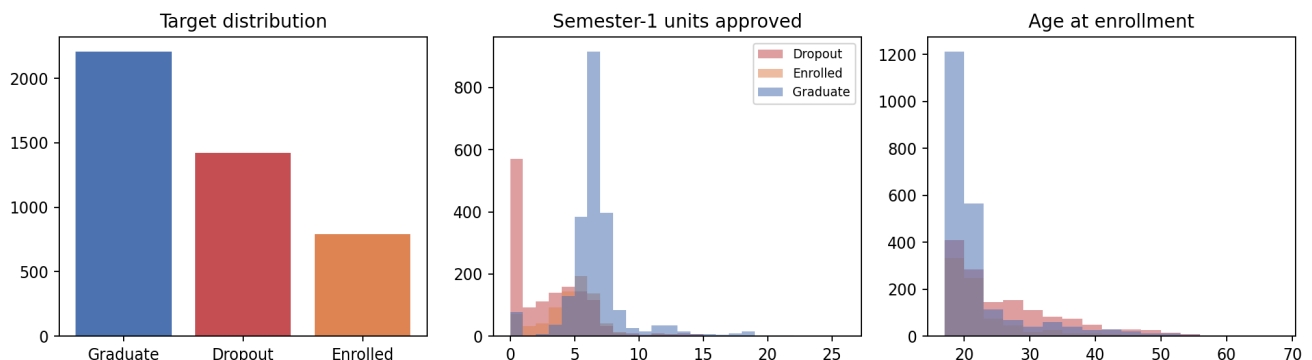
Name / source	Predict Students' Dropout and Academic Success, UCI Machine Learning Repository (doi 10.24432/C5MC89); described in Realinho et al. (2022), <i>Data</i> 7(11):146.
License	Listed on the UCI page as CC BY 4.0 (attribution required). Team to re-confirm on the dataset page before submission.
Provenance	Administrative and national databases of one Portuguese higher-education institution; students enrolled in several undergraduate programmes; anonymized (no names or IDs).
Unit of analysis	One student (row). No repeated measurements.
Size and types	4,424 rows; 36 features (29 integer, 7 float) + 1 text target. 9 features are integer-coded categories; others numeric or binary.
Missingness	0 missing cells; 0 duplicate rows. (The source reports it as pre-cleaned.) One header (Daytime/evening attendance) has a trailing tab, handled in code.
Class balance	Graduate 2,209 (49.9%); Dropout 1,421 (32.1%); Enrolled 794 (17.9%).
Known limitations	Single institution and country; label for Enrolled students is not final; collection time of financial variables not documented; some categorical variables have many rare levels (e.g., Father's occupation: 46 levels, 32 with fewer than 10 rows).

Sample inputs and outputs (first four rows, selected features):

Row	Course	Age	Gender	Schol.	Fees OK	Adm. grade	Sem-1 appr.	Sem-2 appr.	Target
0	171	20	1	0	1	127.3	0	0	Dropout
1	9254	19	1	0	0	142.5	6	6	Graduate
2	9070	19	1	0	0	124.8	0	0	Dropout
3	9773	20	0	0	1	119.6	6	5	Graduate

Categorical codes (Course, Application mode) follow the dataset documentation and are not human-readable; the Data Manager will add a lookup table to the repo.

3. Initial exploratory analysis



Key findings (details in the notebook):

- **Academic progress separates the classes.** Mean semester-1 units approved: Dropout 2.6, Enrolled 4.3, Graduate 6.2. The strongest correlations with Dropout are semester-2 grade (-0.57), semester-2 units approved (-0.57), semester-1 grade (-0.48) and tuition up to date (-0.43).
- **Financial variables are strongly associated with outcome:** 86.6% of students whose fees are not up to date dropped out (vs. 24.7% of those up to date); 62.0% of debtors dropped out (vs. 28.3%); scholarship holders dropped out 12.2% of the time (vs. 38.7%).
- **Demographic differences exist:** Dropout rate is 45.1% for gender code 1 vs. 25.1% for code 0 (coding per dataset documentation); 25.7% of students enrol at age 25 or older, and older students drop out more. International students (2.5%) and students with special needs (1.2%) are very small groups, so subgroup estimates for them will be unreliable.
- **Anomaly:** 180 students have zero enrolled units in both semesters (77 Dropout, 75 Graduate, 28 Enrolled), probably credit transfers or recording artefacts. We keep them and test sensitivity without them.

4. Risks and ethics

- **Leakage / timing.** Semester-2 features and possibly the fee/debtor variables may be recorded close to or after the outcome. Mitigation: two declared feature sets (early, full) and a run without financial features; any feature whose timing cannot be justified is reported as a limitation.
- **Sensitive attributes.** Gender, nationality, marital status, special needs and parental background are personal attributes. The data are anonymized and public, but we request instructor confirmation that this satisfies the course rule on sensitive data. We will not present the model for individual decisions, and will report subgroup performance.
- **Ambiguous label.** Enrolled is not a final outcome; we keep three classes as the main task and report a Dropout-vs-not-Dropout view as a robustness check.
- **Overfitting to a popular dataset.** We use no external notebook code and will cite any published results only as context.
- **Generalization.** One institution; conclusions apply to similar settings only.

5. Train / validation / test strategy

Stratified 70/15/15 split with seed 42 (stored in *data/split_indices.json*), keeping the class mix in every part:

Split	Graduate	Dropout	Enrolled	Total
Train	1546	994	556	3096
Validation	332	213	119	664
Test	331	214	119	664

Hyperparameters are tuned by 5-fold stratified CV on the training set; the validation set is used for model checks; the test set is used once at the end. Scaling and one-hot encoding are fit inside pipelines on training data only. Final results use repeated runs/seeds and report variability.

6. Primary metric

Macro-averaged F1 (unweighted mean of per-class F1), fixed now and not to be changed after seeing results. Justification: the classes are imbalanced (a majority-class model gets 50% accuracy but only 0.22 macro-F1), and missing a dropout is more costly than a false alarm, so minority-class performance must count equally. We also report accuracy, per-class precision/recall, and confusion matrices.

7. Feasibility and compute plan

A single default run per model on the validation set (not a tuned or final result) confirms the task is learnable and that the benchmarks are clearly beaten:

Model	Early set macro-F1 / acc	Full set macro-F1 / acc
Majority class	0.222 / 0.500	0.222 / 0.500
Stratified random (seed 42)	0.313 / 0.373	0.313 / 0.373
Logistic regression (default)	0.651 / 0.736	0.705 / 0.773
Decision tree (depth 5)	0.549 / 0.703	0.602 / 0.720

The dataset is small (about 0.5 MB); every model trains in seconds on a laptop, so no GPU or paid resources are needed. Tools: Python 3, pandas, scikit-learn, matplotlib; GitHub for version control with issues, branches and pull requests. Random forest and cross-validated tuning are expected to take minutes at most.

8. Requests for approval and changes since Milestone 1

- Approve the dataset, the 3-class task and macro-F1 as the primary metric.
- Confirm that this anonymized public dataset is acceptable under the sensitive-data rule.
- Clarification: the contract's Milestone 3 text mentions a linear regression model; because our task is classification, our course-level baseline is multinomial logistic regression. Please confirm this is acceptable.
- The proposal's early-versus-full feature comparison is now defined concretely as 30 vs. 36 features.